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PROJECT AND TEAM INFORMATION

Project Title

Campus WiFi Mesh Topology Optimizer

Team Name:	*DSCPP-III-2026-T023*
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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

The Campus WiFi Mesh Topology Optimizer is a project designed to improve campus WiFi network connectivity by optimizing the connections between wireless Access Points (APs). Network parameters such as distance, latency, interference, signal strength, and user load are used to calculate connection costs. Prim's algorithm is applied to find a Minimum Spanning Tree (MST), which connects the Access Points with minimum total calculated cost while reducing unnecessary connections. Dijkstra's algorithm is used to determine the minimum-cost route between selected Access Points. This system is implemented over C using file handling to store network informations.

Updated Project Approach and Architecture

The Campus WiFi Mesh Topology Optimizer, which uses graph theory to represent nodes i.e. WiFi access points and finding efficient connections between them. Data structures are used to store AP information, represent the network, and help algorithms to find optimized connections and routes. System Design and Approach: Our project is divided into 3 Modules: Module 1 : Network Modeling, which accepts AP parameters such as signal strength, latency , bandwidth, interference , and user load.It calculaes AP health score, Network problems , stores data in text file network.txt. Module 2 : Topology Optimization , reads the stored network information , calculates the cost for each conection and applies algorithms to identify lower-cost set of connections while maintaining network connectivity.

Tasks Completed

Task Completed	Team Member
Project topic selection, problem identification, and objective definition.	Siddharth Shridhar
Literature review and study of graph theory and WiFi topology optimization.	Vansh Kakkar
System architecture design and division of the project into three modules.	Anirudh Rana
Module 1: Network modeling, AP health score calculation, problem detection, and data storage in `network.txt`.	Vansh Kakkar, Abhishek Negi
Module 2: Weighted connection cost calculation and topology optimization using Prim's algorithm.	Siddharth Shridhar, Anirudh Rana
Module 3: Route calculation using Dijkstra's algorithm and basic performance analysis.	Vansh Kakkar, Anirudh Rana
Integration planning, testing with sample network data, and project documentation.	All team members

Challenges/Roadblocks

Initially, undersatanding how the APs are reperesented as nodes and distance between them as wieghted edges.After gathering all the network information, we developed a formula to calculate cost. Later with all network information we implemented Data structure algorithms such as Prim's for topology optimization and Dijkastra's for finding best route required. Passing data from one module to another using file handling was also a challenge we faced. we currently passing data by user manually including high user loads , interference, and failed connections. ensuring that the project provides meaningful network optimization rather than only basic graph operations. we added performance comparisons, failure analysis, and improved network health reporting.

Tasks Pending

Task Pending	Team Member
Complete Module 3: Routing and Performance Analysis using Dijkstra's algorithm to find the best route between access points.	Anirudh Rana, Vansh Kakkar
Integrate Module 3 with Modules 1 and 2 and verify that data is transferred correctly through project files.	Vansh Kakkar, Siddharth Shridhar
Perform final testing using different numbers of access points, connection costs, and network scenarios.	Siddharth Shridhar, Abhishek Negi
Debug the program, handle invalid inputs, and check for disconnected networks and unavailable routes.	Abhishek Negi, Anirudh Rana
Verify optimized connections and routing results, and compare original and optimized network costs.	All team members
Prepare final project documentation, screenshots, and presentation for the demonstration.	All team members

Summary:
The remaining work will focus on completing Module 3, integrating all modules, resolving errors, and conducting final testing. The team will verify routing results and optimized network costs to ensure that the project meets its objectives.

Project Outcome/Deliverables

The main outcome after this project it accepts all the access points detailes including signal strength, bandwidth, latency , user load, interference and generates AP health report.For optimization uses data structure agorithms which reducing unnecessary connections and identifies a minimum cost network.Stores information files which can be shared between the modules for smooth play between them.

Progress Overview

The Campus WiFi Mesh Topology Optimizer is currently in a development and integration stage Module 1 and Module 2 have been implemented to handle network modeling, AP health analysis, problem detection and topology optimizations using data structure Prim's algorithm. Module 3, which requires further integration on routing and performance analysis using dijkstra's algorithm. Final integration, debugging , testing with different network scenarios , verification of optimization results.

Codebase

https://github.com/aperionxerex-boop/PBL-SEM-3.git

Testing and Validation Status

Test Type	Status	Notes
Network Modeling (Module 1)	Pass	The program accepts Access Point details, calculates health scores, detects potential network issues, and stores network information in `network.txt`.
Routing and Performance Analysis (Module 3)	Pending	Dijkstra's algorithm and route calculation still require integration and validation with the other project modules.

Deliverables Progress

Key Deliverable	Current Status	Progress Details
Network Modeling Module	Completed	Accepts AP details and calculates health scores.
AP Health and Problem Detection	Completed	Identifies potential network issues using predefined thresholds.
Topology Optimization Module	Completed	Uses Prim's algorithm to select minimum-cost connections.
Network Data File Handling	Completed	Stores and transfers network data using `network.txt` and `optimized.txt`.
Routing and Performance Analysis Module	In Progress	Dijkstra's algorithm requires final integration and validation.
Final Testing and Validation	Pending	Comprehensive testing, debugging, and verification of results remain.

Overall Progress:
The core network modeling and topology optimization deliverables are implemented. The remaining work focuses on completing and integrating Module 3, conducting final testing, and preparing the final project demonstration.